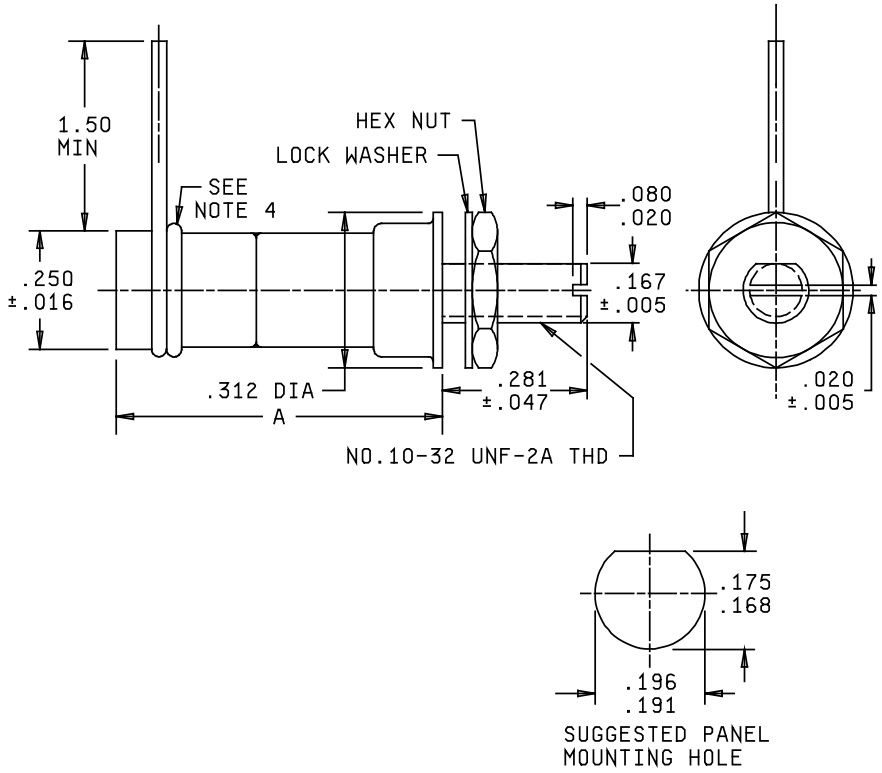
INCH-POUND
MIL-PRF-14409/6F
23 February 2024
SUPERSEDING
MIL-PRF-14409/6E
6 December 2018

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, VARIABLE (PISTON TYPE, TUBULAR TRIMMER), STYLES PC40 AND PC50

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-14409.



Inches	*mm*
.005	0.13
.016	0.41
.020	0.51
.047	1.19
.080	2.03
.167	4.24
.168	4.27
.175	4.45
.191	4.85
.196	4.98
.250	6.35
.281	7.14
.312	7.92
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Unless otherwise specified, tolerance is ± .016 inch.
4. There shall be a minimum of one and one-half turns of wire around the body. These turns shall be completely soldered to the body.
5. The flat on the mounting bushing shall extend to the mounting surface of the shoulder.
6. For dimension A, see table I.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS:

Dimensions: See figure 1 and table I.

DC voltage rating: 750 volts.

Construction: Glass or quartz, see figure 1 and table I.

Insulation resistance (IR): MIL-STD-202-302, test condition B (500 volts ± 10 percent); not less than 1,000,000 megohms, except characteristics J and H shall be not less than 100,000 megohms at +125°C.

Capacitance range: See table I.

Quality factor (Q): See table I.

Thermal shock: MIL-STD-202-107, test condition A, except the step 3 temperature shall be +125°C ± 5°C for characteristics J and H, and +150°C ± 5°C for characteristic Q.

TABLE I. Capacitor characteristics.

Part or identifying number (PIN)	Capacitance range		Dimensions 1/ A +.047, - .031 (+1.19, - 0.79)	Q	Dielectric material	Wire size AWG
	Min	Max				
	pF	pF				
PC40Q1R8	0.6	1.8	.297 (7.54)	1,500	Quartz	24
PC40Q5R5	0.6	5.5	.547 (13.89)	1,500	Quartz	22
PC40Q9R5	0.6	9.5	.984 (24.99)	1,500	Quartz	22
PC40Q160	0.8	16.0	1.594 (40.49)	1,500	Quartz	22
PC50J5R5	0.8	5.5	.297 (7.54)	1,000	Glass	24
PC50J110	0.8	11.0	.547 (13.89)	900	Glass	22
PC50H160	0.8	16.0	.750 (19.05)	800	Glass	22
PC50H230	0.8	23.0	.984 (24.99)	700	Glass	22
PC50H380	1.0	38.0	1.594 (40.49)	500	Glass	22

1/ Metric equivalents are given for general information only. Millimeters are in parentheses.

TABLE II. Substitutability data.

Superseded type designation in MIL-C-14409/6	Substitutable type designation in MIL-PRF-14409/6A through /6F
PC40J4R5 and PC46J050	PC50J5R5
PC40J8R5 and PC46J100	PC50J110
PC40H120 and PC46H140	PC50H160
PC40H180 and PC46H190	PC50H230
PC40H300	PC50H380

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Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents: In addition to MIL-PRF-14409, this document also references:

MIL-STD-202-107
MIL-STD-202-302

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2024-017)

Review activities:
Navy - AS, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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